

SDIE VERSION 6 - STRUCTURAL REASONING ENGINE (REVISED PROMPT)

Derived from: SDIE Version 5 - DeepSeek Structural Reasoning Engine.docx

Implementation baseline: P2_Slabversion5 @ b3ee8c0, P2_SlabVersion6 layer-discovery fork

Last updated: June 2026 (Revised Inizio Project Knowledge - Tagged files corpus)

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MISSION (unchanged from V5 docx)

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Redesign SDIE into a Structural Reasoning Engine that interprets structural drawings the way an experienced structural estimator does.

V4 foundations: Component Atlas, Layer Profiles, DeepSeek ambiguity resolution, Structural Graphs, Knowledge Base.

V5 adds: estimator-style multi-evidence reasoning, topology-enriched DeepSeek prompts, review queue, semantic building model.

V5/V6 implementation adds: slab intelligence with gated exclusions, estimator bay merge, multi-panel floor zones, Inizio podium grid, project-aware soft-layer arbitration.

V6 adds: drawing-driven layer discovery (no hard-coded S_FRAMES default on Inizio sheets).

V6 (Revised Knowledge): INIZIO teaching corpus is reorganized under Revised Project Knowledge/Tagged files/. The model must learn how the estimator draws closed polylines on the raw structural plan and assigns component tags (Slab, Beam, Column, Shear Wall) matching the four supervised tagged references. All downstream pipeline steps are unchanged.

The objective is structural understanding first; slab quantities are deterministic and run only after classification + exclusions.

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PRIMARY OBJECTIVE

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Increase component classification accuracy from 50-60% toward 85-95%.

Primary target classes: Slab, Beam, Column, Shear Wall

Supporting: Structural Wall, Lift Core, Stair Core, Shaft, Opening, Unknown

Business calculators (deterministic, post-classification):

Slab, Beam, Column, Shear Wall - all other types support exclusions/diagnostics.

INIZIO learning objective (Revised Project Knowledge):

Input → Raw structural DXF (framelines, annotations - no estimator tags)

Teach → Four tagged DXFs (estimator-drawn polylines + component labels)

Output → On raw file: infer the same slab bays, beam runs, column footprints, and shear-wall regions the estimator tagged in the teach set

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REVISED INIZIO PROJECT KNOWLEDGE BASE (AUTHORITATIVE - JUNE 2026)

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Root: P2_SlabVersion6/Revised Project Knowledge/

ONLY the Tagged files folder is the project knowledge / teaching corpus for INIZIO. Do not use legacy paths (Data Source/Project1 - Inizio/, old manifest DXF names, or V4/V5 teach folders) when building atlas, layer profiles, KB, or component ground truth for INIZIO.

■■ TEACHING CORPUS (supervised - estimator tags) ■■

Revised Project Knowledge/Tagged files/

Component	File
Slab	Inizio Slab with tag_Revised1.dxf
Beam	Inizio - Beam with tag_Revised1.dxf
Column	Inizio - Colum with tag_Revised1.dxf
Shear Wall	Inizio - Shearwall with tag_Revised1.dxf

Each tagged file is a single-component teaching sheet: every entity on that drawing represents the estimator's answer for that business type. The model must learn:

- Where closed LWPOLYLINE (and supporting LINE/HATCH) regions are drawn
- How slab bays are bounded vs beam centerlines vs column footprints vs shear-wall envelopes
- Which layers carry framing (S-BEAM, S-BEAM-IDEN, A-FLOR-IDEN, S-COLS, ...)
- How annotations (*THK, beam marks, floor IDs) anchor quantity regions

Tagged corpus statistics (Revised1):

Slab teach: ~14.9k entities, ~2.3k closed LWPOLYLINE (slab polygons)
 Beam teach: ~18.9k entities, beam linework + IDs
 Column teach: ~967 entities, column footprints (S-COLS, compact polys)
 Shearwall teach: ~3.4k entities, wall polylines/lines on S-BEAM/S-COLS

■■ INFERENCE TARGET (unsupervised – raw plan) ■■

Revised Project Knowledge/Raw File/

Inizio - Slab beam_Raw_Revised1.dxf

The raw file is the production input: structural geometry and annotations without estimator component polylines. The pipeline must:

1. Read raw geometry using V6 layer discovery + floor zone (unchanged)
2. Classify entities using knowledge learned from Tagged files only
3. Reconstruct framed regions equivalent to estimator slab polylines
4. Tag/classify beams, columns, shear walls as in the four teach sheets
5. Run exclusions → slab intelligence → deterministic quantities (unchanged)

Raw file statistics (Revised1):

~17.5k entities, ~1.4k closed LWPOLYLINE (unlabeled – not GT polys)
 Primary frame layer: S-BEAM (~5.5k entities)

■■ WHAT CHANGES vs LEGACY INIZIO TEACH ■■

REPLACE:

Data Source/Project1 - Inizio/Inizio Slab with tag.dxf
 Data Source/Project1 - Inizio/Inizio - Beam with tag.dxf
 Data Source/Project1 - Inizio/Inizio - Colum with tag.dxf
 Data Source/Project1 - Inizio/Inizio - Shearwall with tag.dxf

WITH:

Revised Project Knowledge/Tagged files/*_Revised1.dxf (four files above)

projects_manifest.json INIZIO entries MUST point to Revised Project Knowledge paths when rebuilding teach artifacts (atlas, layer_profiles, structural_kb).

■■ TEACHING WORKFLOW (scripts – paths updated for Revised corpus) ■■

1. build_layer_profiles.py → data/layer_profiles.json (INIZIO rules)
2. ingest_estimator_workbooks / teach flow → atlas + KB from Tagged files only
3. evaluate_component_classification.py --project-id INIZIO --tagged-only
 GT source: four Revised1 tagged DXFs (not legacy Project1 folder)

Success criterion for INIZIO teach:

Classifier + atlas retrieve patterns from Revised1 tagged polylines/layers such that raw-file inference reproduces estimator-style component tagging before slab quantity engine runs.

■■ CURSOR / AGENT CONTEXT (INIZIO) ■■

@Revised Project Knowledge/Tagged files/

```
@Revised Project Knowledge/Raw File/  
@data/layer_profiles.json  
@data/atlas/component_atlas.json  
@data/knowledge_base/structural_kb.json  
@src/sdie/ingestion/layer_discovery.py  
@src/sdie/classification/  
@src/sdie/detection/
```

Do NOT @-reference legacy Project1 - Inizio DXFs for new INIZIO work.

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PIPELINE ORDER (IMPLEMENTED – UNCHANGED STEPS)

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1. Load DXF + resolve units/scale (src/sdie/ingestion/dxf_reader.py)
2. Layer discovery (V6) OR CLI override (src/sdie/ingestion/layer_discovery.py)
3. Floor zone from *THK label clusters + frame extent (floor_zone.py)
4. Extract entities from resolved structural + annotation layers
5. Classify: rules → layer profiles → geometry → atlas → graph → DeepSeek (v5)
6. Build structural graph (graph/engine.py)
7. Slab intelligence: exclusions → beam grid / label merge / podium grid
8. Deterministic quantities (quantity/slab.py)
9. Outputs: results.json, building_model.json, overlay.html, review_queue.json

Entry point: scripts/run_pipeline.py

```
--auto-layers (default) | --no-auto-layers | --layers <override>  
--project-id INIZIO | TRUST_OFFICE | MANOHAR  
--no-deepseek for rule+topology-only runs
```

INIZIO inference example (raw Revised1 plan):

```
python scripts/run_pipeline.py ^  
  "Revised Project Knowledge/Raw File/Inizio - Slab beam_Raw_Revised1.dxf" ^  
  -o Output/Inizio_Raw_Revised1 --project-id INIZIO --no-deepseek
```

Regression / layer-test drawing (separate from teach corpus):

Data Source/TestInput/Inizio_B2_LayerTest_V5.dxf

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LAYER DISCOVERY ENGINE (V6 – NEW)

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Problem (V5): CLI defaulted to --layers S_FRAMES STR-CUTOUT, which returns zero slabs on Inizio drawings that use S-BEAM for both beams and perimeter walls.

Approach: single modelspace scan + project profile boost before classification.

Module: src/sdie/ingestion/layer_discovery.py

Algorithm:

1. scan_layer_stats(msp) – per-layer counts of LINE, LWPOLYLINE, TEXT, HATCH
2. Score structural candidates:
 - lines × 3 + polylines × 2 + hatches
 - × project layer_profiles boost for (project_id, layer)
 - × 1.8 if layer name matches BEAM|FRAME|STR-BEAM|S-BEAM pattern
 - down-rank cutout/annotation-dominant layers
3. Keep candidates ≥ max(25% of top score, 30)
4. Annotation layers: any layer with TEXT/MTEXT or ANNO|IDEN|FLOR|THK in name
5. Cutout layers: STR-CUTOUT, SUNK SLAB, S-SUNKEN HATCH, name match CUTOUT|VOID
6. frame_layers: top 1-2 line-heavy structural layers (for beam-grid + floor zone)

Resolved sets written to PipelineConfig and logged in results JSON as layer_discovery.

Validated auto-discovery:

```
INIZIO → frame_layers: [S-BEAM], structural: [S-BEAM, SUNK SLAB]  
TRUST_OFFICE → frame_layers: [S_FRAMES], structural: [S_FRAMES, STR-CUTOUT, S-SUNKEN HATCH]
```

CLI override: --layers S-BEAM --no-auto-layers (disables discovery for framing)

Diagnostic: scripts/analyze_dxf.py --project-id <ID> prints v6_layer_plan

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CLASSIFICATION STACK (V4 + V5)
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Order of precedence:

1. Void/THK/beam-tag annotation rules
2. Soft-layer geometry arbitration (see below)
3. Per-project layer profiles (data/layer_profiles.json)
4. Hard global layer hints (S-COLS, STR-CUTOUT, A-FLOR-IDEN, ...)
5. Geometry classify (beam_enclosure, column_compactness, wall_continuity)
6. Atlas k-NN vote (layer-indexed)
7. DeepSeek v5 for ambiguous entities only (threshold 75%)

Soft layers (geometry required – never hard-map layer name alone):
S_FRAMES, STR-BEAM, S-BEAM, S-BEAM-IDEN

Soft-layer arbitration (INIZIO S-BEAM – IMPLEMENTED):

If geometry:beam_line with conf $\geq 0.88 \rightarrow$ Beam (interior grid lines)
Elif layer_profile Shear Wall with conf $\geq 0.85 \rightarrow$ Shear Wall (perimeter)
Elif geometry wall_line / wall_polyline $\geq 0.82 \rightarrow$ Shear Wall
Else fall through to project profile rule

Evidence tags for downstream slab exclusions:

layer_profile:<layer>:<entity_type>
geometry:beam_line | geometry:wall_line | geometry:wall_polyline
(NOT geometry:wall – must match classifier output exactly)

V5 multi-evidence weights (src/sdie/confidence/scorer.py):

Layer 25% | Geometry 25% | Topology 20% | Annotation 15% | Atlas 10% | KB 5%
Thresholds: auto_accept ≥ 90 , review ≥ 75 , force_queue < 60

Note: blended V5 scores understate high layer-profile confidence (~89% rule $\rightarrow$ ~56% final). Exclusion gating uses effective confidence floor for profile-backed frame-layer shear walls (see Semantic Exclusions).

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PROMPT ENGINEERING FRAMEWORK (V5 DEEPSEEK – from docx)
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Every DeepSeek request MUST include:

Scale Context – never assume; pass 1:50 / 1:100 / 1:200 explicitly
Unit Context – Millimeters / Meters / Imperial; geometry depends on units
Layer Context – all layer names found + resolved frame_layers from discovery
Geometry Context – entity type, area, perimeter, aspect ratio, bbox,
connected beams/columns/walls, neighbour relationships
Annotation Context – nearby THK, B12, C5, SW1, LIFT, STAIR, OPENING, SHAFT
Cursor/RAG Context – Revised Project Knowledge/Tagged files (INIZIO),
layer_profiles, atlas, KB, config, classifier

DeepSeek system role (verbatim intent):

Senior structural QS / estimator / CAD specialist
Classify only – forbidden from area, volume, concrete, shuttering
Return confidence + flag uncertain entities for review

FEW-SHOT ESTIMATOR REASONING (INIZIO – Revised Tagged files)

Before classifying entities on the RAW plan, retrieve similar patterns from the four Revised1 tagged teach sheets (not legacy Project1 DXFs).

Retrieve:

Top 5 similar Atlas samples (built from Tagged files/ only)
Top 5 similar Knowledge Base samples (INIZIO Revised corpus)

Teach file → business type mapping:

Inizio Slab with tag_Revised1.dxf → Slab (closed bay polylines)
Inizio - Beam with tag_Revised1.dxf → Beam
Inizio - Colum with tag_Revised1.dxf → Column
Inizio - Shearwall with tag_Revised1.dxf → Shear Wall

Inference mapping on raw plan:

Long narrow S-BEAM LINE → match Beam teach linework
Closed region in beam grid → match Slab teach polylines (*THK bays)
Compact S-COLS / hatch → match Column teach footprints
Continuous S-BEAM wall poly → match Shearwall teach envelopes

Use nearest Revised1 estimator examples before DeepSeek classification.

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FLOOR ZONE ENGINE (multi-panel DXF – IMPLEMENTED)

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Module: src/sdie/detection/floor_zone.py

Purpose: stacked plans on one sheet (tower + basement satellite + duplicate copy).

Method: thk_cluster on A-FLOR-IDEN *THK labels

- Primary cluster = largest THK band (B2 tower on Inizio)
- Satellite clusters merged only if within gap of primary (not far-below stacks)
- Exclude Basement-03 tags far below primary ($y < \text{primary.y_min} - 0.35 \times \text{gap}$)
- Subpanel split when duplicate tower copy detected (internal Y gap)
- grid_bounds_y capped at selected subpanel ymax
- frame ymin extends floor bounds to lowest S-BEAM/frame line (podium margin)

Outputs: bounds_y (grid), label_bounds_y (THK filter), notes in results JSON

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SLAB INTELLIGENCE ENGINE v3.3 (IMPLEMENTED)

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Module: src/sdie/detection/slab_intelligence.py

Never identify slabs first. Classify → exclude → derive framed bays.

Strategy selection (priority):

1. semantic_label_merged_bay – *THK labels merged when $\geq \text{min_thk_labels_for_merge}$
2. semantic_beam_grid_bay[_merged] – orthogonal beam centerline grid
3. +podium_grid – supplementary bays below THK band (Inizio podium)

Config highlights (src/sdie/config.py):

min_slab_area_m2: 0.4
merge_beam_grid_to_estimator_bays: true
apply_slab_exclusions: true
use_layer_gated_semantic_exclusions: true
enable_podium_beam_grid_slabs: true
frame_layer_shear_wall_exclusion_min_confidence: 55

Beam grid (detection/beam_grid.py):

- Collect orthogonal LINE axes on frame_layers
- Line intersects Y band (not centroid-only) for vertical beams in podium band
- augment_sparse_grid_axes: add band edges + primary X cluster for sparse grids
- ExclusionCatalog difference per cell before area threshold

Label merge (detection/slab_by_label.py):

- Merge *THK TEXT/MTEXT into estimator bays using grid_cells as seeds
- bounds_y = label_bounds_y from floor zone

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ESTIMATOR BAY MERGE (IMPLEMENTED – Trust Office S43 fix)

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Module: src/sdie/detection/bay_merge.py

Merges adjacent raw beam-grid cells into physical estimator bays.

Key behaviours:

- infer_from_cells: auto span/area caps from cell population
- Column-stack merge: vertical neighbours sharing X column (i fixed, j±1)
- _span_touch_or_overlap: allow merge when grid lines touch/overlap (not only identical edge coordinates) – fixes 1 m sliver bays blocking 27 m² merge
- _column_stack_pair / _is_row_sliver: stack thin row slivers without bridging full adjacent bays
- Elevated area caps only for column-stack merges (not global)
- auto_inferred=True preserves infer_from_cells caps when merging

Regression: tests/test_bay_merge.py (includes S43-style vertical stack)

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PODIUM BEAM GRID (INIZIO – IMPLEMENTED)

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Band: ymin = floor_bounds_y[0] to ymax = thk_label_bounds_y[0] - gap_mm

_infer_podium_bounds_y:

- Do NOT collapse ymax to tight horizontal-beam cap when framing is sparse (Inizio corner detail at y=1-3 m below full 15 m band).
- Use ymax_candidate when horizontal span < 12% of band height.

Sparse grid helpers (beam_grid.py):

- _select_primary_vertical_axis_cluster: drop duplicate plan X copies
- _augment_horizontal_axes / _augment_vertical_axes for 1-2 axis grids
- min_vertical_span uses Y-band overlap length (tower walls crossing band)

Typical Inizio result: 2 podium cells ~7.7 m² at low Y + tower label-merge bays

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SEMANTIC EXCLUSION GATING (IMPLEMENTED)

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Module: slab_intelligence._classified_exclusion_geoms + _layer_allows_semantic_exclusion

Geometric exclusions: columns, hatches, void labels, cutouts (layer lists in config)

Semantic exclusions (classified components):

Gated by layer + confidence + evidence

Frame-layer shear walls on S-BEAM (Inizio):

Allow exclusion when:

- component_type Shear Wall or Structural Wall
- layer in frame_layers
- evidence starts with layer_profile: OR geometry:wall_line OR geometry:wall_polyline
- effective confidence ≥ frame_layer_shear_wall_exclusion_min_confidence (55)
OR layer_profile boost to 88 for gating (V5 blend understates rule conf)

Do NOT deduct interior beam centerlines on frame_layers (Beam classification).

Buffer widths: wall 200 mm, column 50 mm, core 4500 mm, opening 1500 mm

Results JSON fields:

semantic_exclusion_area_m2, semantic_exclusion_parts, total_exclusion_area_m2

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STRUCTURAL EXCLUSION REASONING (docx + implementation)

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Before slab identification, classify and exclude:
Beam, Column, Shear Wall, Structural Wall, Lift Core, Stair Core, Shaft, Opening

Remaining framed region = potential slab.
No slab quantity before exclusion analysis completes.

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REVIEW QUEUE (V5)
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Output: *_review_queue.json
Entities with confidence < 75% (or force_queue < 60%)
Fields: entity_id, classification, confidence, evidence, alternatives

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VALIDATION & BENCHMARKS (June 2026)
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INIZIO component teach / eval (Revised Project Knowledge):
  scripts/evaluate_component_classification.py --project-id INIZIO --tagged-only
  GT: four files under Revised Project Knowledge/Tagged files/*_Revised1.dxf
  Each file = 100% supervised type (slab / beam / column / shearwall sheet)

INIZIO raw inference validation:
  Run pipeline on Raw File/Inizio - Slab beam_Raw_Revised1.dxf
  Compare overlay polylines + classified components against Tagged files teach
  patterns (visual + entity-level precision/recall vs teach geometry statistics)
  Slab quantity GT (when available): data/ground_truth/Inizio Slab with tag.json

Other projects (unchanged):
  TRUST_OFFICE / MANOHAR – legacy manifest + Project2/3 folders until revised

Slab quantity (workbook):
  scripts/verify_test_gt.py
  GT: data/ground_truth/*.json or TestGT Excel

Reference runs (P2_Slabversion5 – LayerTest regression, not Revised teach):
  Trust Office FF: ~27/30 GT bays matched (terrace run), S43 27.54 m² bay fixed
  Inizio B2 LayerTest: 70 slabs, 2336 m² (workbook 2045 m²)

Overlay: src/sdie/validation/overlay.py – layer toggles, GT match coloring,
  compare raw inference output against Revised1 tagged teach overlays

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HISTORICAL PROJECT LEARNING (UPDATED)
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INIZIO (authoritative corpus):
  Revised Project Knowledge/Tagged files/ – ONLY source for teach rebuild
  project_id: INIZIO

TRUST_OFFICE / MANOHAR:
  Data Source/projects_manifest.json + Project2/3 folders (unchanged until
  revised knowledge packs are added)

Every INIZIO teach rebuild contributes from Revised1 tagged files:
  Atlas samples (closed polylines, linework, layer+geometry features)
  Layer profiles (project_id=INIZIO, layer, entity_type rules)
  Structural KB patterns (annotation + topology + bay framing)
  Component GT for evaluate_component_classification.py

Knowledge accumulates in data/ – source DXFs remain in Revised Project Knowledge/.

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KEY MODULE MAP (P2_SlabVersion6)

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```
=====
src/sdie/
  ingestion/      dxf_reader, entity_extractor, layer_discovery (V6), units
  classification/ classifier, rag_classifier, layer_profiles, features, types
  detection/      slab_intelligence, beam_grid, bay_merge, floor_zone,
                  exclusions, slab_by_label, region, beam_frame
  graph/          engine.py
  model/          building.py
  confidence/     scorer.py (V4 + V5 weights)
  reasoning/      v5_context, deepseek_client, component_classification
  validation/     overlay, gt_match, component_gt, component_eval, review_queue
  semantic_pipeline.py, pipeline.py, config.py

Revised Project Knowledge/ (INIZIO – authoritative teach + raw inference)
  Tagged files/          four *_Revised1.dxf component teach sheets
  Raw File/              Inizio - Slab beam_Raw_Revised1.dxf (inference input)
```

```
data/ (generated from Tagged files – not hand-edited for INIZIO teach)
  atlas/component_atlas.json
  knowledge_base/structural_kb.json
  layer_profiles.json
  ground_truth/Inizio Slab with tag.json (workbook totals – slab BOQ check)
```

```
scripts/
  run_pipeline.py, analyze_dxf.py, build_layer_profiles.py,
  evaluate_component_classification.py, verify_test_gt.py, diagnose_s43.py,
  teach_all_projects.py (must use Revised Project Knowledge paths for INIZIO)
```

```
=====
CLASSIFICATION TARGETS (aspirational)
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```
Beam / Column / Shear Wall / Slab accuracy  $\geq$  90%
Overall classification  $\geq$  90%
Low-confidence entities  $\leq$  5%
```

```
=====
FINAL DIRECTIVE (from V5 docx – still applies)
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```
Do not improve slab detection by geometry hacks alone.
Improve structural reasoning and teach estimator patterns.
```

For INIZIO: learn polyline-and-tag behaviour from Revised Project Knowledge/ Tagged files/, then apply the same structural understanding to the Raw File plan. The tagged sheets are the estimator's ground truth for HOW to bound and label components – not optional reference material.

DeepSeek = structural reasoning engine.
Quantity engine = deterministic.

All quantities only after:

```
Teach corpus load (Tagged files) → Raw plan load → Layer discovery →
Component Classification → Structural Graph → Semantic Building Model →
Structural Exclusion Analysis → Slab Intelligence → Quantity Engine
```

The objective is structural understanding.